

Optical classification and Dynamics Study of the OH Megamasers Galaxy IRAS 11506-3851

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ABSTRACT

We present an optical spectroscopy study of OH Megamaser Galaxy (OHMG) IRAS11506-3851 using Gemini/GMOS and images taken from Hubble Space Telescope. Our HST i band image of IRAS11506-3751 shows numerous compact region of ongoing circum-nuclear star formation. One-dimensional spectra of the OHMG show [OIII]5007, [OI]6300, of $H_{\alpha}\lambda$ 6564,61 and [NII] $\lambda\lambda$ 6549,86 6585,27 , and [SII]6717,31 emission lines, and a very intense $H\beta$, and NaD absorption line which may indicate young stellar cluster, and neutral gas clouds of the ISM , respectively. Through the rotation curve obtained through the 158 deg slot we have the amplitude is approximately 95 km/s. WHAN diagnostic diagram have been used to investigate the dominant excitation mechanisms present in the IRAS11506-3751 system host an strong-AGNs in its center and circum-nuclear region with shock ionization may be due to the interaction and/or supernovae from the stellar cluster.

I - INTRODUCTION

OH Megamaser Galaxies (OHMGs) from a sub-class accounting for approximately 20% of [Ultra] Luminous Infrared Galaxies ([U]LIRGs), which emit OH maser lines at 1665 and 1667 MHz with a brightness approximately $10^{2-4}L_{\odot}$ (Lo, 2005). As these galaxies are found in gas-rich mergers, it has been suggested that OHMs can be used to trace galaxy merger rates and associated processes (dust obscured star formation and black hole growth) over a wide redshift range (Lo, 2015, and references therein). Frequently, the OH maser emission is found in galaxies with high concentrations of dense, $n(H_2) = 10^{5-7}cm^{-3}$ molecular gas, sometimes in edge-on rotating disks or rings on scales of <100pc. In addition, the OH lines often show broad asymmetric profiles and velocity shifts suggestive of outflows (Baan, 1989, Rovilos, 2002, Parra, 2005, Momjian, 2006). Keeping this in mind, OHMGs may therefore represent a critical, short-lived transition phase in which massive, dense concentrations of molecular gas are triggering intense episodes of star formation and the onset of AGN fueling, resulting in rapid black-hole growth.

One outstanding question is whether starburst or AGNs are the dominant sources of dust and gas heating, hence, the main goal of this study is investigating the dominant excitation mechanisms, and dynamics of the OHMG IRAS 11506-3851. Through the i band image taken with the HST (see Figure 1), the source IRAS 11506-3851 is a regular isolated disk with numerous compact regions of circum-nuclear stellar formation in progress, whose main basic properties are summarized in Table 1.



Figure 1: Image of IRAS 11506-3851 taken by HST.

Table 1: General Properties of IRAS 11506-3851.

Properties	Value
z	0.01078
D _L	46.6
Scale (pc/arcsec)	222
Inclination angle	(37±3)°

II – OBSERVATION AND DATA REDUCTION

The optical spectroscopy observations are obtained by Gemini South (GS) telescope, through projects GS-2013B-Q-90 (PI: Dinalva A. Sales) and GS-2019A-Q-90 (PI: Andrew). The data were taken from Gemini Multi-Object Spectrograph (GMOS) in the long-slit mode with a 1'' and 0.75'' slit width. The IRAS 11506-3851 was observed using GS telescope, and the P.A. of the three long-slit is demonstrated in Figure 2. In this case, we can observe nucleus as well as circum-nuclear star forming regions. The PA of 261deg are taken using a slit with 1'', and the other PA are taken using the slit with 0.75''.

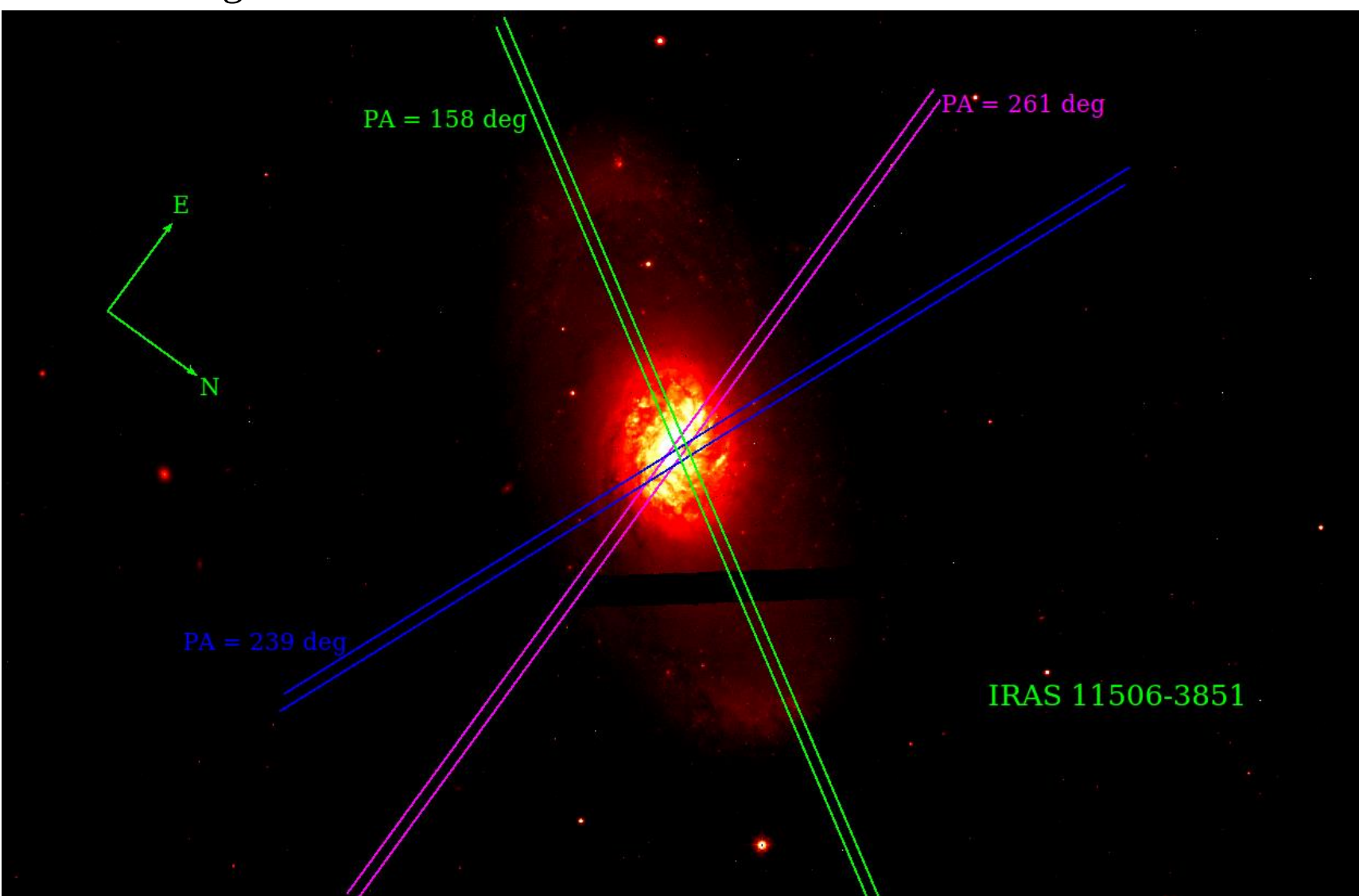


Figure 2: Images of IRAS11506-3851 composed by SDSS 4680A (in red), HST F814W (in green, i band) and HST Hα+[NII] (Sales et al. in prep.).

The spectroscopic data reduction have been done using the software IRAF, along with the packages developed for the data reduction of Gemini telescopes, specifically for the GMOS instrument. The data reduction process followed the standard steps used for long-slit data reduction. The description of the tasks can be found in GEMINI.GMOS.

III – DATA ANALYSIS

Figure 3 shows spectra, in the wavelength range between roughly 4500Å and 7200Å, of IRAS 11506-3851 taken from PA 261deg (pink spectrum) in two different regions the circum-nuclear star forming region north-West (see figure 3-a) and the nuclear spectrum (see figure 3-b). Figure 4 show the nuclear spectrum of the galaxy taken from PA 158deg (green spectrum), and figure 5 show the nuclear spectrum of the galaxy taken from PA 239deg (blue spectrum). In all spectrum clearly show $H_{\alpha}\lambda$ 6564,61 and [NII] $\lambda\lambda$ 6549,86 6585,27 emission lines, also, I cannot see emission lines of [OIII] λ 5007, [OI] λ 6300, and [SII] λ 6717,31.

We also observed in the spectrum of IRAS 11506-3851 NaD absorption doublet lines, at where the existence of two global components is expected as the NaD absorption feature may have originated in the neutral gas clouds of the ISM as well as in the stars.

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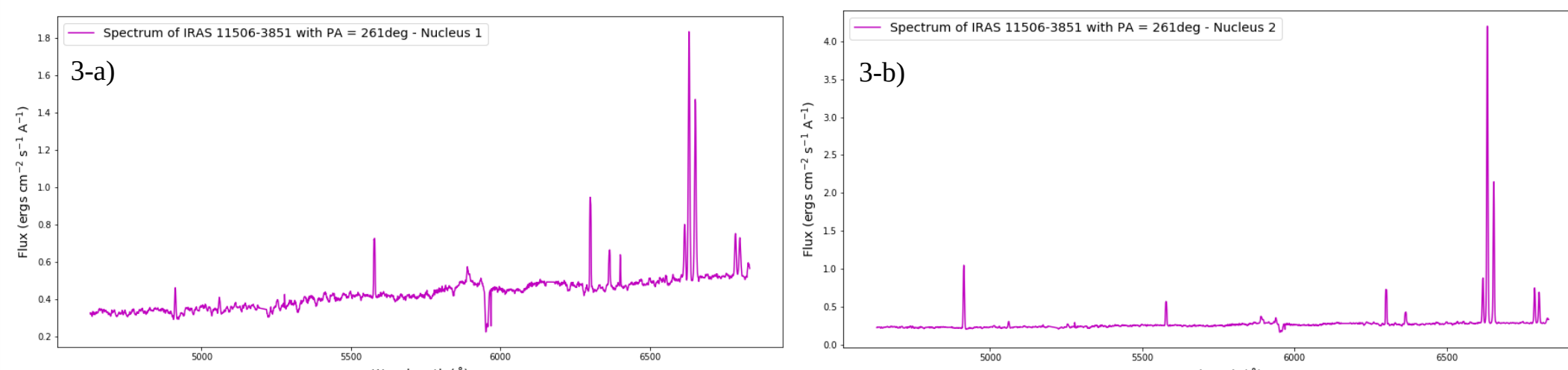


Figure 3: One-dimensional spectra of the IRAS 11506-3851 taken by PA of 26deg. Left-panel (3-a): spectrum of a circum-nuclear star forming region North-West of the nucleus. Right-panel (3-b): spectrum of the nucleus of the galaxy.

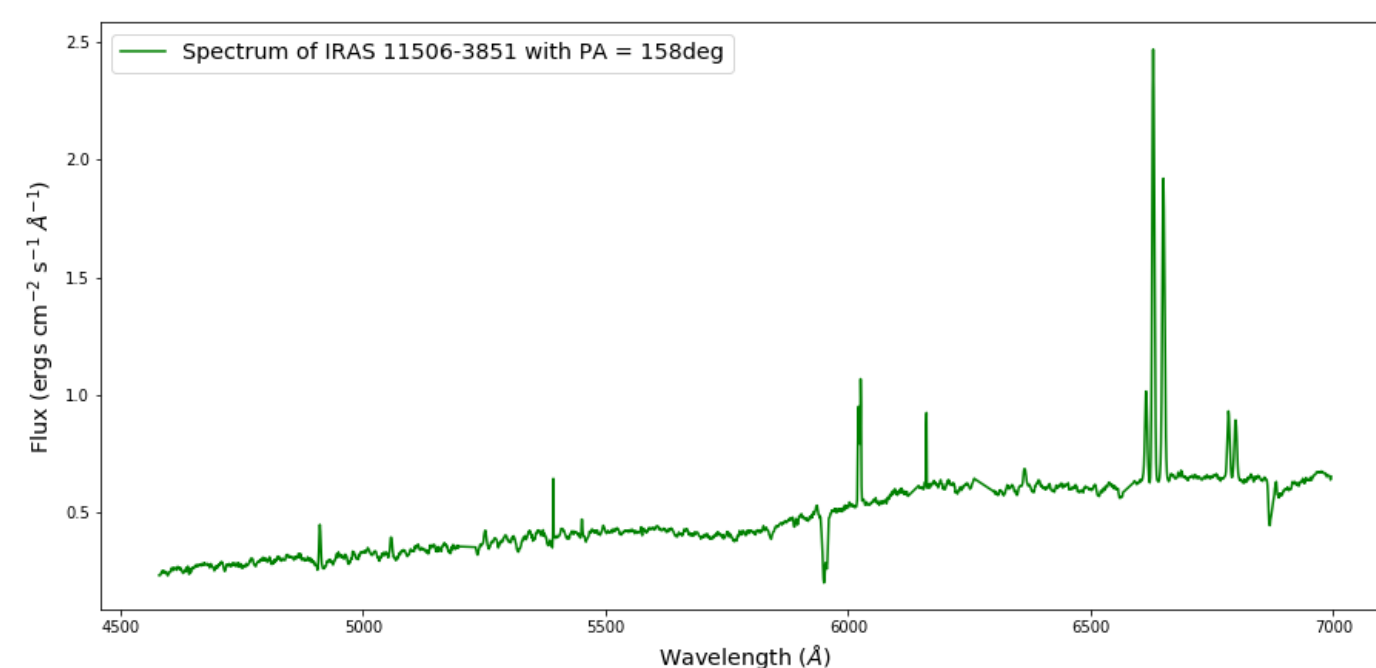


Figure 4: One-dimensional spectra of the IRAS 11506-3851 taken by PA of 158deg.

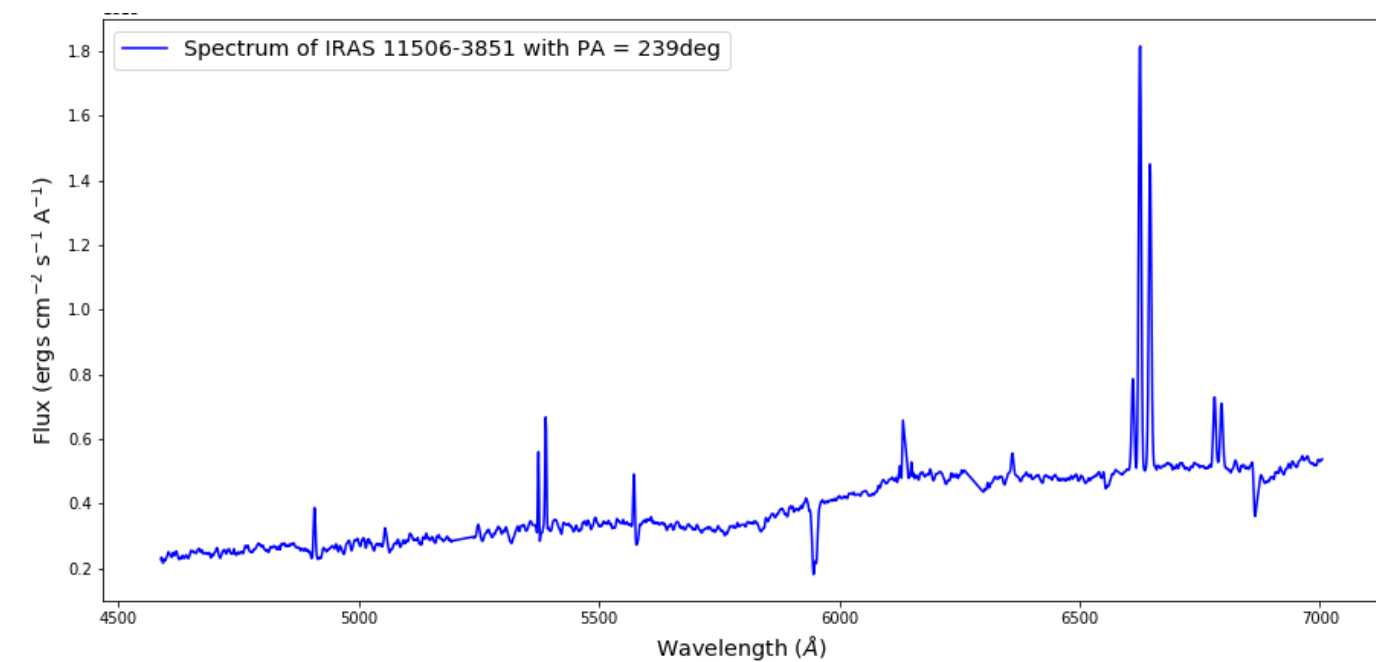


Figure 5: One-dimensional spectra of the IRAS 11506-3851 taken by PA of 239deg.

III.1 – IONIZED GAS KINEMATICS

We derived the rotational curve of the galaxy using the H_{α} emission line along the three slits. Figure 6 show the rotational curve of the galaxy taken from PA 261deg (pink curve), figure 7 show the rotational curve of the galaxy taken from PA 239deg (blue curve), and figure 8 show the rotational curve of the galaxy taken from PA 158deg (green curve). The figure 9 illustrates the displacement of the H_{α} line along the PA 158deg slit, where the green spectrum is the center of the galaxy. Through the graphs of the rotation curves we can see that the slit with PA158deg is close to the major axis of the galaxy, and the slits with PA 239 and 261deg are close to the minor axis of the galaxy.

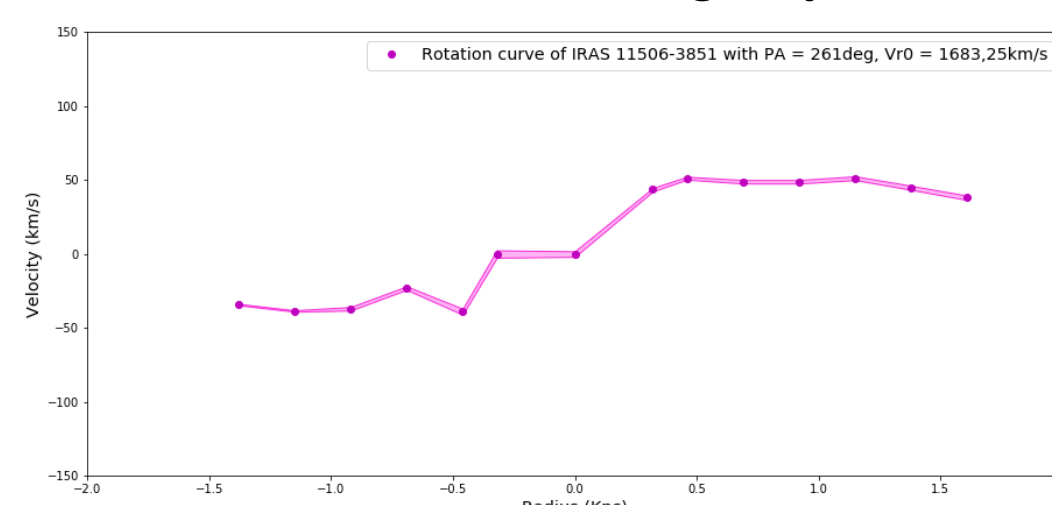


Figure 6: Rotational curve of the IRAS 11506-3851 through slit with PA of 261deg.

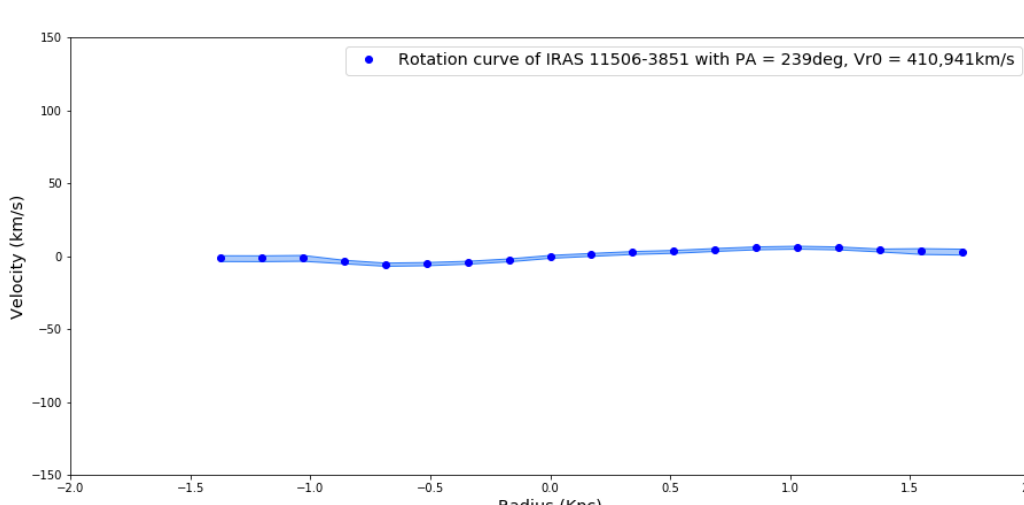


Figure 7: Rotational curve of the IRAS 11506-3851 through slit with PA of 239deg.

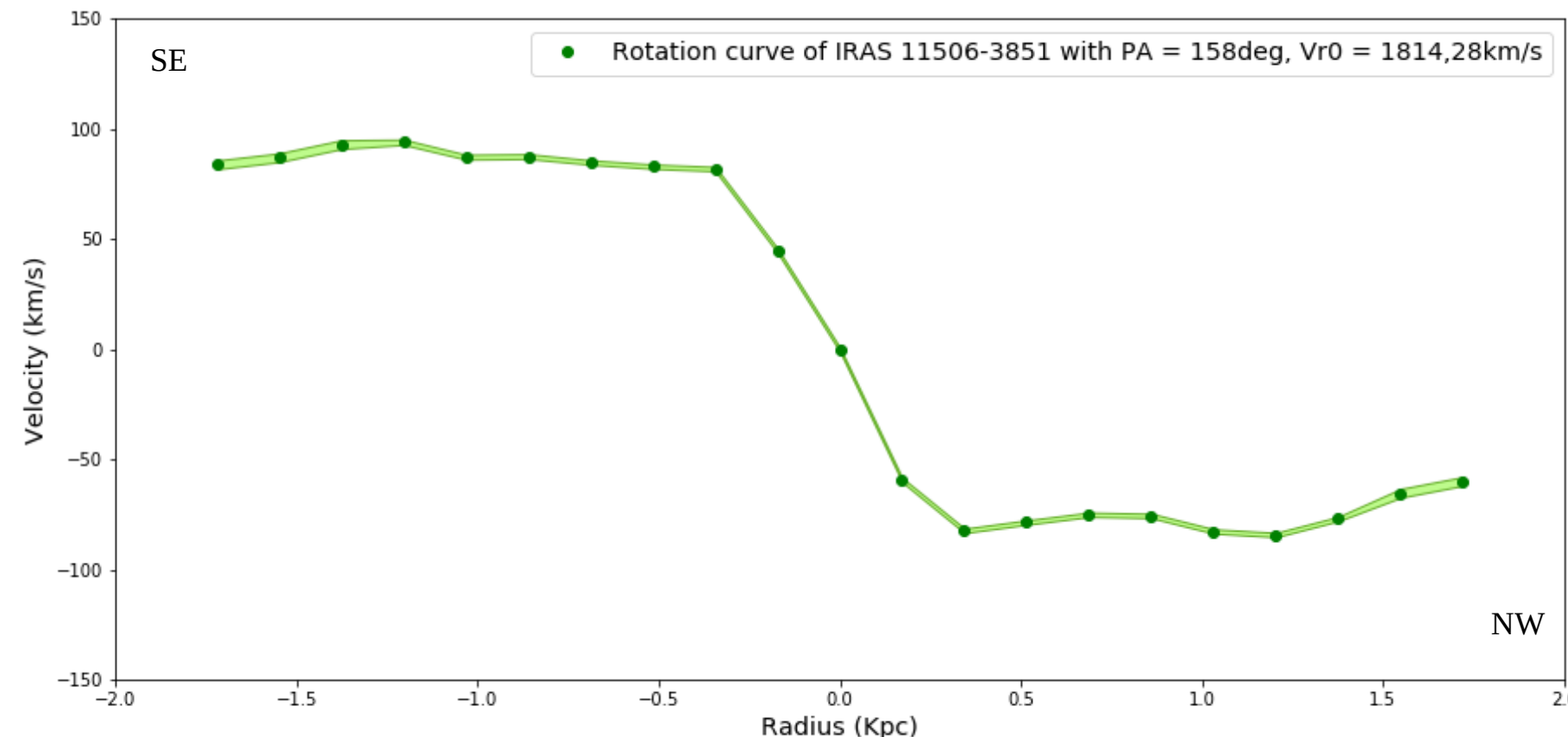


Figure 8: Rotational curve of the IRAS 11506-3851 through slit with PA of 158deg.

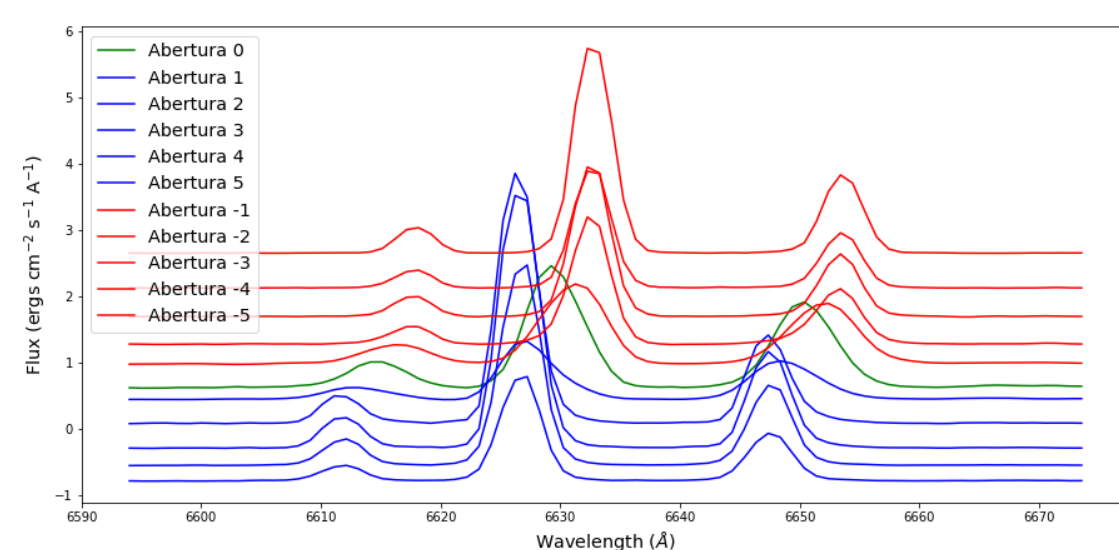


Figure 9: displacement of $H_{\alpha}\lambda$ 6564,61 and [NII] $\lambda\lambda$ 6549,86 6585,27 emission line along the slit with PA 158deg..

III.2 – EXCITATION MECHANISMS OF IRAS 11506-3851

We derived fluxes, and equivalent width (EWs) of the H_{α} and [NII]6583 emission lines present in the one-dimensional spectra of IRAS11506-3851, in order to construct the WHAN diagnostic diagram (Cid Fernandes et al., 2011). First, the WHAN diagram was obtained using the central spectrum of all slits (spectra of Figures 3, 4 and 5), see Figure 10. In addition, Figure 11 shows the WHAN diagram which was constructed using spectral data along the with PA 158deg, where the extractions are shown in figure 12.

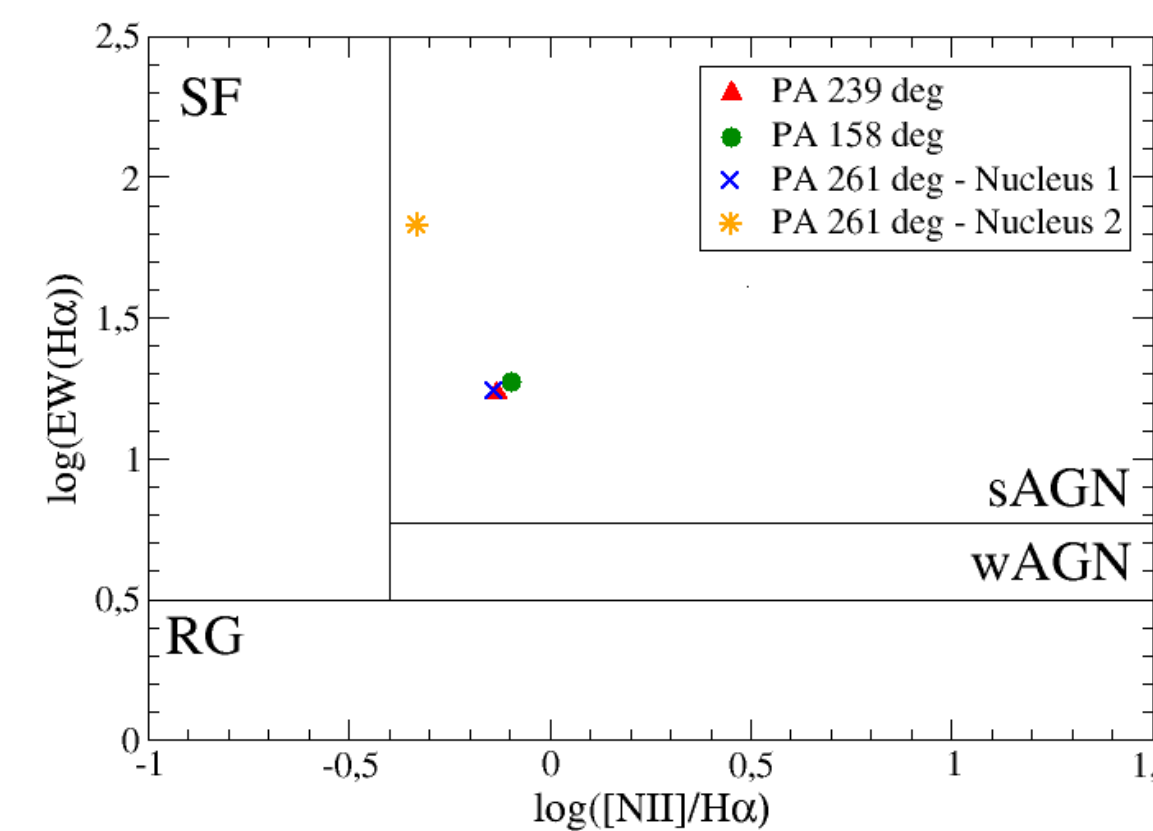


Figure 10: WHAN diagnostic diagram of IRAS11506-3751 taken by the nuclear spectrum of all slits. The SF, sAGN, wAGN and RGs are delimited regions that correspond respectively of the star-formation, weak-AGN, strong-AGNs and retired galaxies.

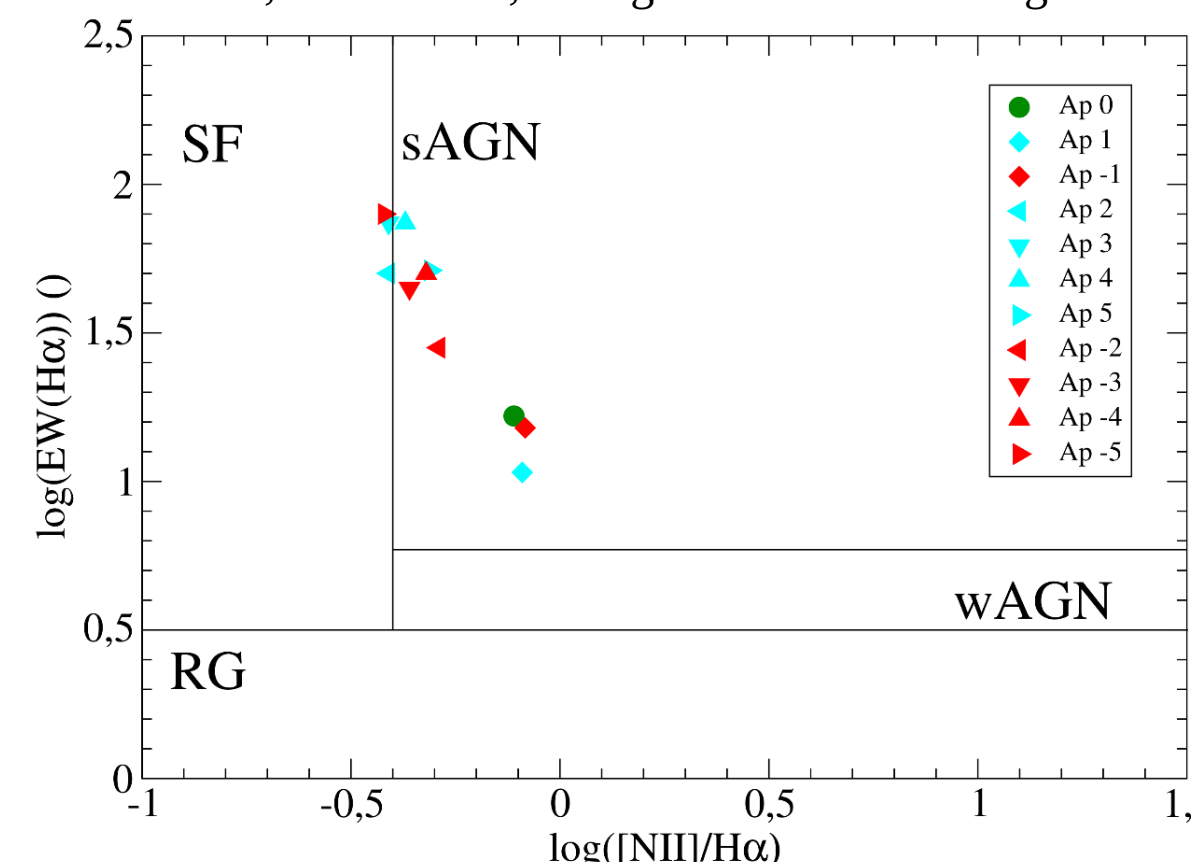


Figure 11: WHAN diagnostic diagram of IRAS11506-3751 taken by along of all slit with PA 158deg.

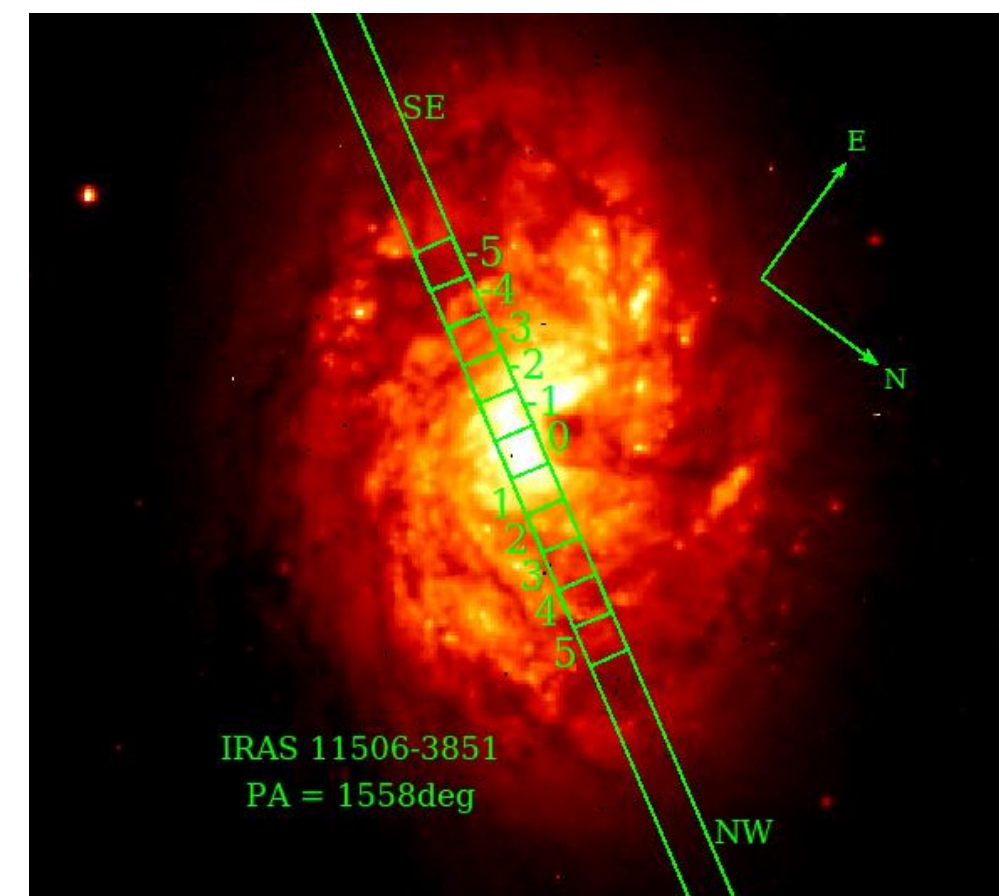


Figure 12: Image of IRAS11506-3751 and the slit with PA 158deg and demonstrating the positions of the extractions along the slit.

IV – CONCLUSION

From this work we conclude:

- From figure 6, 7 and 8 we could derive from corrected rotation curve amplitudes of approximately 50 km/s, 10 km/s, and 95 km/s for PAs of 261deg, 239deg, and 158deg, respectively.. The IRAS 11506-3851 mass distribution model is under study, and will be modelled as the sum of the bulge, disc and halo components.
- From figure 10, we could conclude that all PA of IRAS11506-3851 hosts in its nucleus a strong AGN, whereas the circum-nuclear region seems also to be consistent with photoionization by the sAGN, however more investigation need to be addressed in order to compare whether this shock ionization is driven by either sAGN and starburst super winds or galaxy collisions.
- From Figure 11, we can conclude that in the central region of the slit with PA158 deg is photoionized through an sAGN, but in the slit exterminations the regions is located in the limit between star-forming objects and sAGN.

In the future we will still extract the stellar or gas kinematics and stellar population of the galaxy spectra by a full spectrum fitting using the pPXF software. We will also investigate the BPT diagnostic diagram for comparative purposes.

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